TECHNISCHE UNIVERSITÄT CHEMNITZ

Department of Electrical Engineering and Information Technology

Chair of Measurement and Sensor Technology

Project Documentation

„Project Lab Embedded Systems“

Group: 17

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Project: Zero Potential Circuit Optimization for Cross-Talk Elimination in Matrix Sensor Systems.

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1 Table of Contents

2 Abstract

Two-dimensional (2-D) sensor arrays are widely used but suffer from crosstalk, a phenomenon where parasitic current paths degrade measurement accuracy. Traditional DC-based methods like the Zero Potential Circuit (ZPC) attempt to mitigate this issue but are often compromised by non-ideal components. This report presents a comprehensive analysis, simulation, and implementation of an advanced measurement system based on the Alternating Current Zero Potential Circuit (AC-ZPC) methodology. The AC-ZPC approach utilizes frequency-division multiplexing, driving each sensor row with a unique frequency to enable simultaneous, independent, and crosstalk-free measurements.

The proposed sensor interface circuit was rigorously validated through automated LTspice simulations. An AC analysis established an operational frequency limit of 13 kHz for high signal integrity, while extensive transient analyses were performed using a variety of excitation signals, including single-sine, multi-sine, square wave, linear chirp, and DIBS. These simulations consistently demonstrated exceptionally high accuracy, with measurement deviations typically remaining below 0.08%, confirming the robustness of the frequency-domain methodology even with complex broadband inputs. Furthermore, a novel column-based calibration technique was introduced to compensate for real-world non-idealities such as row interface impedance.

Finally, a hardware prototype was successfully implemented using an STM32H7B0VBTx microcontroller, demonstrating a complete, high-performance solution for real-time impedance spectroscopy. The findings confirm that the AC-ZPC method, combined with

robust data processing and a practical calibration strategy, provides a reliable and highly accurate solution for advanced sensor imaging applications, effectively overcoming the limitations of conventional techniques.

3 Member Responsibilities

Mohammed Hesham	Hardware design and source code
Eyad Gawish	Documentation
Ibrahim Hassan	Simulation
Abdelrahman Mansour	Final Presentation

4 Functional Description

4.1 Overview

4.1.1 Hardware Setup

The physical setup consists of the main measurement device, which houses the STM32 microcontroller and associated electronics, and the sensor matrix you wish to test.

- **Connect the Sensor Matrix:** The primary connection you need to make is to attach your 2D sensor matrix to the designated input ports on the measurement device. These ports link the rows and columns of your sensor matrix to the system's internal multiplexers.
- **Power the Device:** Connect the main measurement device to a suitable power source as specified by the hardware design.
- **Connect to a Host PC:** Use a UART (Universal Asynchronous Receiver-Transmitter) cable or a USB-to-UART adapter to connect the measurement device to a host computer. This connection is essential for viewing the measurement results.

4.1.2 System Operation

Once the hardware is connected, the system is designed for straightforward operation. The complex tasks of signal generation, data acquisition, and processing are handled automatically by the microcontroller.

- **Initiate a Scan:** The system is designed to perform a full scan of the connected sensor matrix. Upon powering up or receiving a start command from the host PC (depending on the specific software interface), the device will begin the measurement process.
- **Automated Measurement:** The system will automatically:
  1. Inject a predefined AC signal into each row of the sensor matrix.
  2. Select the corresponding column to measure the output.
  3. Repeat this process iteratively until every sensor in the matrix has been measured.
- **View the Results:** The final, processed impedance data is transmitted to the connected host PC. You will need a data logging or visualization program (such as a serial terminal or a custom-built application) running on the PC to receive and display the data. The output will be an "impedance map"—a 2D array of values that mirrors the

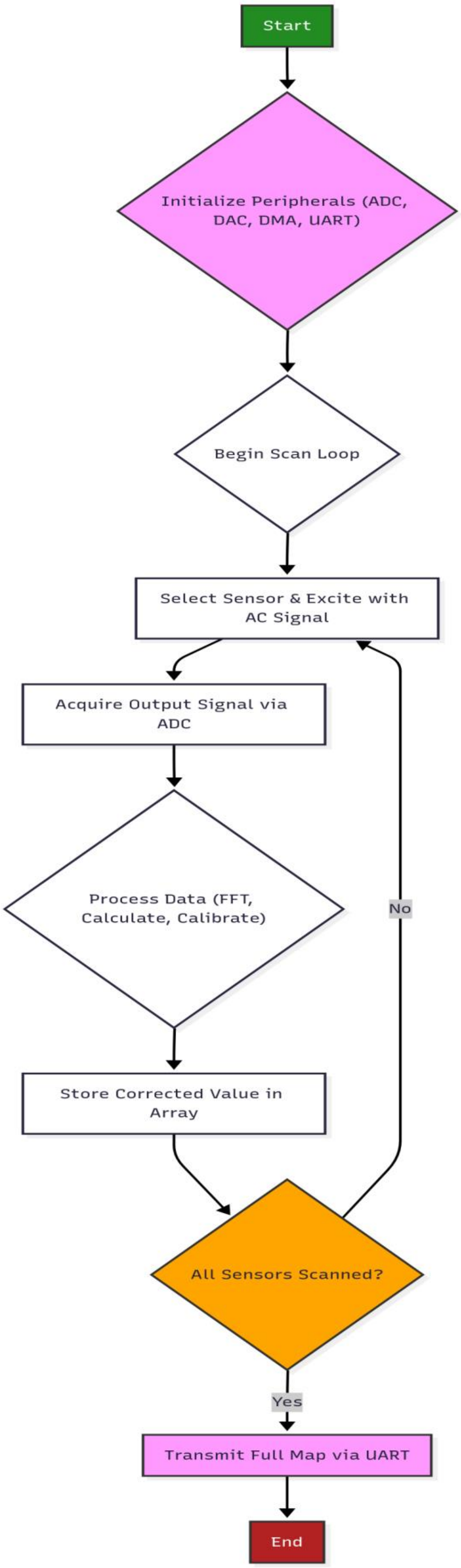
physical layout of your sensor matrix, showing the measured impedance for each individual sensor. This allows for a complete visualization and analysis of the sensor array's state.

The system is designed to be largely autonomous. Your primary role as a user is to connect the sensor matrix and the PC, and then use a software interface on the PC to view the final impedance map.

4.2 Software

This flowchart outlines the main operational loop of the microcontroller's software for scanning the sensor matrix.

1. **Start:** The process begins.
2. **Initialize Peripherals:** The first step is to configure all necessary hardware modules on the microcontroller. This includes the Analog-to-Digital Converter (ADC), the Digital-to-Analog Converter (DAC), the Direct Memory Access (DMA) controller, and the UART communication interface.
3. **Begin Scan Loop:** The software enters the main loop to start the process of scanning the sensor matrix.
4. **Select Sensor & Excite with AC Signal:** Within the loop, the system first selects a specific sensor in the matrix using the multiplexers. It then uses the DAC to generate an AC signal to excite that sensor.
5. **Acquire Output Signal via ADC:** The ADC is used to capture the resulting output signal from the selected sensor.
6. **Process Data:** The captured data is then processed. This involves performing a Fast Fourier Transform (FFT), calculating the impedance based on the FFT results, and applying a calibration function to correct the value.
7. **Store Corrected Value in Array:** The final, accurate impedance value is stored in a 2D array in the microcontroller's memory.
8. **All Sensors Scanned?** A decision is made. The system checks if every sensor in the matrix has been measured.
    - If **No**, the loop returns to step 4 to select and measure the next sensor.
    - If **Yes**, the process moves to the next step.
9. **Transmit Full Map via UART:** Once all sensors have been scanned and their values stored, the complete impedance map (the 2D array) is transmitted to the host computer.
10. **End:** The process is complete.


Start
Initialize Peripherals (ADC, DAC, DMA, UART)
Begin Scan Loop
Select Sensor & Excite with AC Signal
Acquire Output Signal via ADC
Process Data (FFT, Calculate, Calibrate)
Store Corrected Value in Array
All Sensors Scanned?
No
Yes
Transmit Full Map via UART
End

4.3 Hardware

Hardware System Description

This document provides a summary of the hardware developed for a high-performance, real-time impedance measurement system for 2D sensor arrays, covering key requirements, design choices, and verification results.

Application Requirements

The hardware was developed to perform accurate, real-time impedance spectroscopy on 2D sensor arrays. Key requirements included eliminating sensor crosstalk, ensuring robustness against non-ideal components (like op-amp offsets), supporting various AC excitation signals for full impedance spectroscopy, and having the flexibility to scan an entire sensor matrix to generate a complete impedance map.

Design Process

The design is based on the **Alternating Current Zero Potential Circuit (AC-ZPC)** method, which uses frequency-division multiplexing to prevent crosstalk. The core of the system is a powerful **STM32H7B0VBTx microcontroller**, chosen for its integrated high-speed ADC/DAC, DMA for efficient data handling, and an FPU for hardware-accelerated FFT computations. The periphery includes external analog multiplexers for sensor selection and high-quality operational amplifiers (ADA4522) to ensure signal integrity.

Simulations

The design was validated using extensive, automated LTspice simulations. An AC analysis determined a maximum operational frequency of **13 kHz** to maintain signal integrity. Transient analysis, using a wide variety of excitation signals (including single-sine, multi-sine, square wave, chirp, and DIBS), confirmed the system's high accuracy, with measurement deviations consistently remaining **below 0.08%**.

Verification Measurements

A physical hardware prototype was successfully built and tested. The system demonstrated its ability to perform the entire measurement pipeline from signal generation to FFT processing in real-time. A software-based polynomial calibration function was implemented to compensate for physical non-idealities, ensuring maximum accuracy. The final verified system successfully scans a full sensor matrix and transmits a complete, calibrated impedance map to a host PC via UART.

5 Description of files

Describe the provided documents within this project containing

- Folder structure
- Filenames + content

project_documentation.docx project documentation

schedule_presentation.docx	schedule presentation slides
final_presentation.pptx	final presentation slides
additional_files	
>diagrams	diagrams and visualisations
>layouts	schematics and circuit diagrams
>source_code	source code for Arduino IDE
>>documentation	doxygen source code documentation

6 References

- Z. Hu, W. Tan, and O. Kanoun, "High Accuracy and Simultaneous Scanning AC Measurement Approach for Two-Dimensional Resistive Sensor Arrays," *IEEE Sensors Journal*, vol. 19, no. 12, pp. 4623-4628, 2019.

- Z. Hu, R. Ramalingame, A. Y. Kallel, F. Wendler, Z. Fang, and O. Kanoun, "Calibration of an AC Zero Potential Circuit for Two-Dimensional Impedimetric Sensor Matrices," *IEEE Sensors Journal*, vol. 20, no. 9, pp. 5019-5025, 2020.

- J. Wu, L. Wang, and J. Li, "Design and Crosstalk Error Analysis of the Circuit for the 2-D Networked Resistive Sensor Array," *IEEE Sensors Journal*, vol. 15, no. 2, pp. 1020-1026, 2015.

- A. Fischer, A. Y. Kallel, and O. Kanoun, "Comparative Study of Excitation Signals for Microcontroller-based EIS Measurement on Li-Ion Batteries," in *2021 International Workshop on Impedance Spectroscopy (IWIS)*, 2021, pp. 44-47.

1 Introduction

Up-to-date sensing systems are relying upon two-dimensional (2D) matrices to detect different physical signals (heat, pressure, etc). Such matrices have a high spatial resolution and cheap and less complex cabling through widespread sharing of rows and columns among multiple sensing elements. However, one of the main disadvantages of standard 2D sensor arrays is that they are affected by the crosstalk in which electrical signals from nearby sensors affect the accuracy of the target sensor's measurement. This is caused due to interfering current paths formed in the matrix structure, especially when resistive or impedimetric sensors are employed. A well-known method to mitigate crosstalk issue is the Zero Potential Circuit (ZPC) based on operational amplifiers to perform a virtual ground for unnecessary elements. This method is extremely efficient, but on the other hand suffering from hardware issues, such as amplifier offset voltages and the fact that physical components such as multiplexers are not ideal, traditional ZPC methods impose significant measurement errors. In order to solve these inaccuracies in measuring,This paper introduces methods called Direct current zero potential Potential Circuit (DC-ZPC) and Alternating Current Zero Potential Circuit (AC-ZPC). By using each row of the matrix being driven by a sine wave of a distinct frequency and looking at the frequency domain output, the method makes it possible to simultaneously, independently, and accurately measure all the sensor elements. Furthermore, the paper shows a calibration method that further corrects for deviations in row interface impedance, which is essential for overall system accuracy.

2 2-D Resistive Sensor Arrays

Two-dimensional (2-D) networked resistive sensor arrays are utilized in a wide range of applications, including tactile sensing, temperature sensing, gas detection, and medical imaging. Their architecture involves sharing rows and columns, which serves to simplify the complexity of interconnections. For a matrix with M rows and N columns, this structure reduces the required number of interconnect lines to M+N.

The fundamental structure of a 2-D sensor array is a grid of resistive elements, where each element is located at the intersection of a row and a column. In an ideal measurement circuit, a specific element, referred to as the element being tested (EBT), can be measured by applying a known voltage and observing the output.

For an ideal simplified measurement circuit, a precise voltage (V_I) is applied across the EBT (R_{xy}) and a known sample resistor (R_s) connected in series. The voltage across the sample resistor, V_{sg}, is measured. The equivalent resistance of the EBT, denoted as R_{sg}, can then be determined. The relationship is given by the voltage divider rule:

$$V_{sg} = \frac{R_s}{R_s + R_{sg}} V_I \tag{1}$$

Since V_I and R_s are known and V_{sg} is measured, the resistance of the EBT (R_{sg}) can be calculated.[3]

3 A Key Challenge in Resistive Sensor Arrays

Despite the simplified wiring, the shared-line structure of 2-D arrays introduces a significant challenge known as crosstalk. Crosstalk occurs when unintended parasitic current paths are formed through un-targeted sensors during the measurement of a targeted sensor. This phenomenon adversely affects measurement performance and leads to systematic measurement deviations.

For instance, as illustrated in Figure 1, when attempting to measure the resistance of sensor R_{11}, the applied voltage can also create a secondary current path (dashed line) through neighboring sensors, such as R_{12}, R_{22}, and R_{21} in parallel with the target sensor. The measured current is therefore a sum of the current through the target sensor and the current through these parasitic paths, causing an error in the calculated resistance. The primary factors influencing crosstalk are the resistance of the multiplexers, the resistance of adjacent row and column elements, and the overall size of the array.[2]

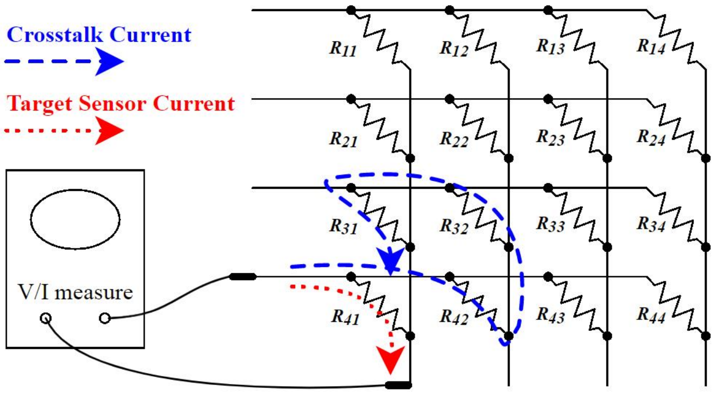


Figure 1: An example of crosstalk in a 2-D sensor array. The measurement of the target sensor (solid current path) is contaminated by a parasitic current flowing through un-targeted sensors (dashed current path).[2]

In the Improved Isolated Drive Feedback Circuit (IIDFC), a DC method designed to mitigate this issue, the equivalent resistance R_{sg} of the EBT (R_{xy}) is influenced by the multiplexer channel resistors for the column (R_c) and row (R_r), and all other elements in the same row as the EBT (R_{my}). The mathematical expression is:

$$R_{sg} = R_c + R_r + R_{xy}\left(1 + \sum_{m=1,m\neq x}^{M} \frac{R_c}{R_{my}}\right) \tag{2}$$

This equation shows that even in an improved DC circuit, the measurement is still affected by adjacent row elements.[3]

4 AC Zero Potential Circuit (AC-ZPC)

4.1 Overcoming DC Circuit Limitations

Two-dimensional resistive sensor arrays are powerful tools used in a multitude of applications, but their measurement accuracy is often hindered by crosstalk. Traditional measurement methods, such as the DC Zero Potential Circuit (ZPC), are widely implemented to reduce this crosstalk effect. However, the accuracy of these DC methods is strongly influenced by non-ideal characteristics of their components. Key sources of measurement deviation include the switch-on resistance of the row-controlling multiplexers and the inherent DC offset voltage of the operational amplifiers (op-amps) used in the circuit. To address these fundamental limitations, the AC Zero Potential Circuit (AC-ZPC) was proposed. This approach leverages AC measurement principles to create a more robust and accurate system that is insensitive to the DC offset voltage of amplifiers and eliminates the need for a row-controlling multiplexer entirely.

4.2 Design and Working Principle

The core innovation of the AC-ZPC is the use of frequency-division multiplexing to isolate each sensor element for measurement, a design based on several key principles. First, for row-specific frequency excitation, each row of the sensor matrix is driven by an AC voltage source operating at a unique frequency. This allows for simultaneous scanning, as each row's signal is distinguishable in the frequency domain, enabling all sensors in the matrix to be measured at the same time a significant improvement over the sequential scanning required by DC methods. Furthermore, each column line is connected to an AC voltage summing circuit, formed by the inverting terminal of an op-amp with a feedback resistor. The output of each column amplifier is therefore a multi-sine wave containing a superposition of the frequency components from every row's excitation signal. Finally, through frequency spectrum analysis, such as a Fast Fourier Transform (FFT), the value of an individual sensor is determined by isolating the response at the unique frequency corresponding to that sensor's row. This design fundamentally changes the measurement approach, making the concept of short-circuiting untargeted sensors, which is central to the DC-ZPC, no longer necessary. Because the measurement of each sensor is separated in the frequency domain, the AC-ZPC method is inherently free from the crosstalk effects that plague DC circuits.[2]

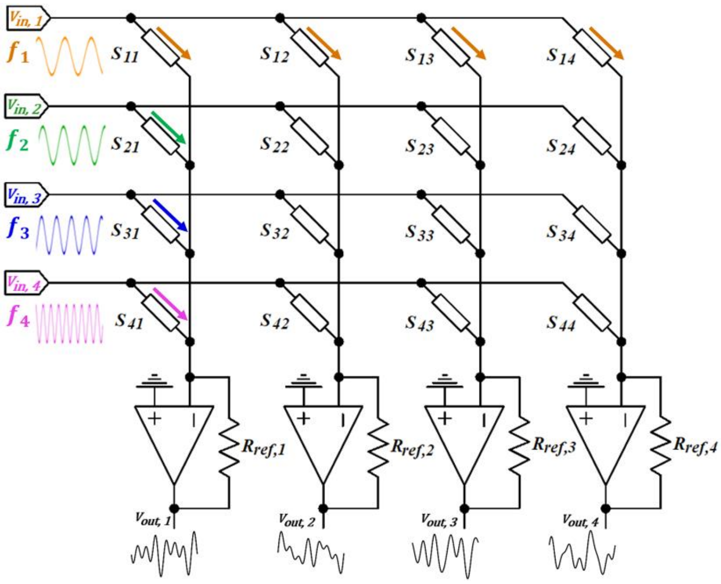


Figure 2: Illustration of the AC-ZPC method for a resistive sensor matrix of four rows and four columns. where each row is driven by a single-sine voltage at a unique frequency. The column outputs are multi-sine signals that are analyzed to determine individual sensor values.[2]

4.3 Governing Equations and System Modeling

For a purely resistive sensor matrix, the relationship between the applied and measured signals is straight forward. The AC voltage source driving the m^{th} row can be described as a sine wave:

$$AC_m = A_m \sin(\omega_m t + \varphi_m) \tag{3}$$

where A_m is the amplitude, ω_m is the unique angular frequency for that row, and φ_m is the initial phase.

The output of the n^{th} column amplifier is a multi-sine wave superposition of the responses from all M rows:

$$V_{out,n} = -\sum_{m=1}^{M} B_{mn} \sin(\omega_m t + \varphi_m) \tag{4}$$

where B_{mn} is the measured amplitude of the output signal at frequency ω_m. Based on the principles of an AC voltage summing circuit, the resistance of the sensor at the intersection of the m^{th} row and n^{th} column (R_{mn}) can be calculated using the known feedback resistor (R_{ref}) and the ratio of the measured output amplitude to the known input amplitude:

$$R_{mn} = R_{ref} \frac{A_m}{B_{mn}} \tag{5}$$

The AC-ZPC method's true potential is realized in its application to impedimetric sensor matrices, where sensors are more complex than simple resistors and are often modeled as a resistor in parallel with a capacitor. The impedance of the sensor Z_{mn} is given by:

$$Z_{mn} = R_{mn} || \frac{1}{j\omega_m C_{mn}} = \frac{R_{mn}}{1 + j\omega_m R_{mn} C_{mn}} \tag{6}$$

In the frequency domain, the transfer function relating the column output voltage to the row input voltage at the specific frequency ω_m is determined by the impedance of the sensor Z_{mn} and the feedback resistor $R_{ref,n}$:

$$\frac{V_{out,n}(\omega_m)}{V_{in,m}(\omega_m)} = -\frac{R_{ref,n}}{Z_{mn}} \tag{7}$$

By measuring both the amplitude gain ($\frac{A_{mn}}{A_m}$) and the phase shift ($\Delta\phi = \beta_{mn} - \varphi_m$) between the output and input signals, the unknown resistance R_{mn} and capacitance C_{mn} can be calculated separately:

$$R_{mn} = \frac{R_{ref,n}}{A_{mn}/A_m}\sqrt{1+\tan^2(\Delta\phi)} \tag{8}$$

$$C_{mn} = \frac{\tan(\Delta\phi)}{\omega_m R_{mn}} \tag{9}$$

To avoid the complexities of phase measurement, an alternative "double-sine" signal model can be used, where each row is excited at two different frequencies. This allows the impedance to be calculated using only amplitude measurements, albeit at the cost of increased measurement time or system complexity.[1][2]

4.4 Real-World Challenges and Calibration

In an ideal AC-ZPC circuit, the row driving voltage sources are perfect. However, in real-world implementations, factors such as the output impedance of the voltage source or the use of switches to route signals can introduce an unexpected "row interface impedance" (Z_m) in series with each row line. This is illustrated in Figure 3.

This seemingly small impedance introduces a new form of crosstalk. The measurement of a target sensor becomes dependent not only on its own impedance but also on the impedance of all other sensors in the *same row*. The transfer function at a specific frequency ω_1 (for row 1) is modified as follows:

$$V_{out,1}|_{\omega=\omega_1} = \frac{-R_{ref,1}V_{in,1}}{Z_{11}\left(1 + Z_1\sum_{n=1}^{N}\frac{1}{Z_{1n}}\right)} \tag{10}$$

As shown in Equation 10, the output voltage for the sensor Z_{11} is now affected by the row interface impedance Z_1 and the sum of admittances of all sensors in that row ($\sum\frac{1}{Z_{1n}}$), reintroducing measurement deviation.

Figure 3: The AC-ZPC method showing the real-world problem of row interface impedances ($Z_1, Z_2, \dots$) which cause measurement deviations.

To suppress the measurement deviations caused by row interface impedances, a novel and effective column calibration approach has been proposed. This strategy involves inserting a reference column, populated with known, pre-characterized impedances, to serve as a set of reference sensors. To find the impedance of an unknown sensor, such as Z11, a ratio-based calculation is used where its output is measured along with the output of the known sensor in the same row (e.g., Z14). By taking the ratio of their transfer functions, the common error term—containing the row interface impedance and the sum of admittances—is canceled out, allowing for an accurate measurement. The resulting calibration equation to find the unknown impedance Z_{11} using the known reference Z_{14} is:

$$Z_{11}|_{\omega=\omega_1} = Z_{14}\frac{R_{ref,1}V_{out,4}}{R_{ref,4}V_{out,1}}\bigg|_{\omega=\omega_1} \tag{11}$$

Since every term on the right side of Equation 11 is either known or measured, the true value of the unknown sensor can be accurately calculated, free from the effects of row interface impedance. This calibration approach has been successfully verified through both simulation and hardware prototype implementations, proving its efficacy in creating high-accuracy impedimetric sensor matrix systems.[2]

4.5 Sensor Interface Circuit and AC Analysis

The circuit under investigation is a sensor interface designed to measure changes in a 4x4 matrix of resistive sensors. The schematic of this circuit is presented in Figure 4. In this configuration, each resistor (e.g., R11, R12, etc.) represents a sensor with a nominal resistance of 100 kΩ.

The first row of the sensor matrix is excited by a 1V AC voltage source, V_{row1}, while the remaining rows are connected to a reference voltage, v_{ref}. The output from each of the four columns is fed into a differential amplifier stage. These amplifiers, implemented using the ADA4522 operational amplifier, measure the voltage difference between their respective column and the reference voltage v_{ref}.

A critical design constraint for this system is dictated by the hardware, which requires that the output voltage from any operational amplifier, v_{out}, must not exceed 3.3V to ensure compatibility and prevent damage to subsequent components.

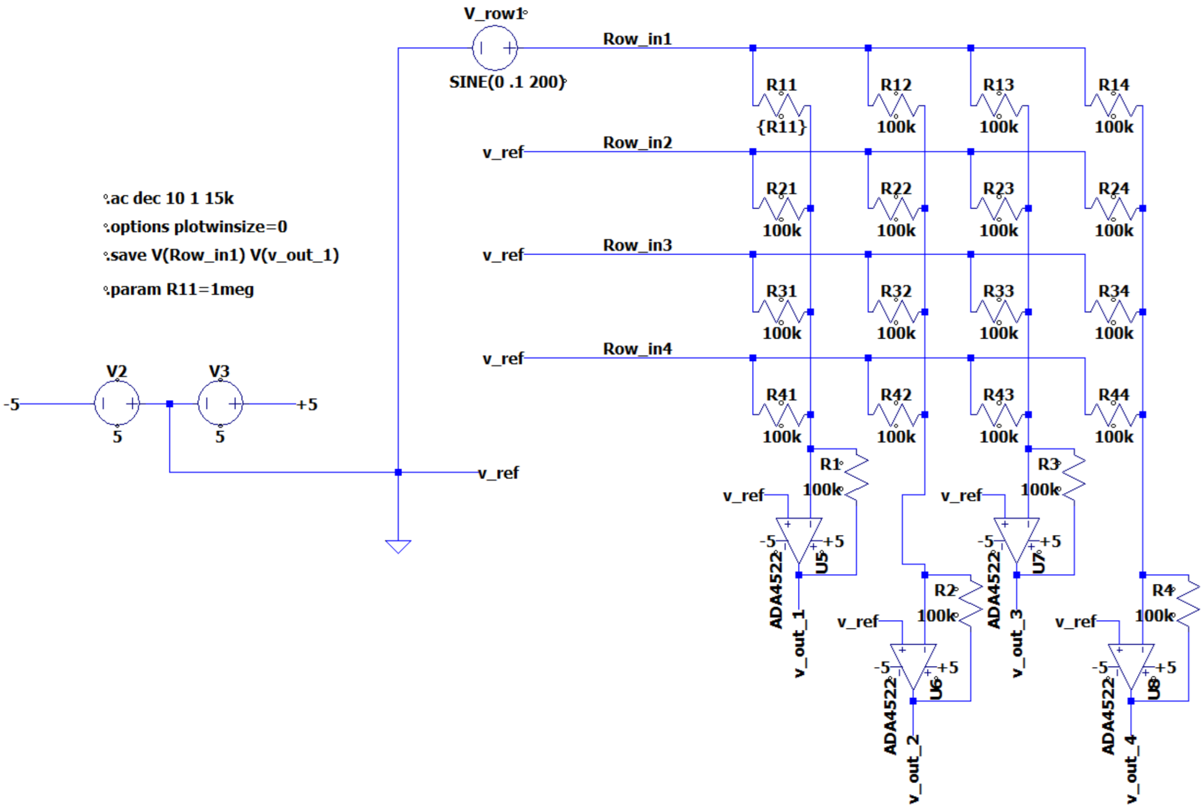


Figure 4: Schematic of the 4x4 resistive sensor matrix interface circuit.

4.6 Simulation and AC Analysis Results

The primary goal of our work is to characterize the frequency response of the circuit through an AC analysis. This involves applying an AC input signal across a range of frequencies and observing the resulting output magnitude and phase.

In an ideal scenario, for a purely resistive network, both the magnitude and phase of the output signal would remain constant regardless of the input frequency. However, the operational amplifiers used in the output stage introduce frequency-dependent behavior. Our analysis focuses on monitoring this deviation, particularly the phase shift. The objective is to identify the maximum operating frequency at which the phase deviation of the output signal remains within a tolerable limit of 2 degrees.

While our current hardware setup utilizes a sensor excitation frequency of 5 kHz, it is crucial to understand the circuit's behavior beyond this specific value. In practical field applications, sensor systems are often required to operate at significantly higher frequencies. Therefore, this simulation was designed to sweep across a broad frequency spectrum to proactively assess the performance limitations of our design.

An AC simulation was performed over a frequency range of 1 Hz to 15 kHz. The simulation was conducted under a specific condition where one sensor's resistance, R11, was changed to 50 kΩ to represent a sensing event. The results for the output of the first channel, v_{out_1}, are shown in Figure 5.

As depicted in Figure 5, the phase of the output signal (red, dashed line) is stable at 180 degrees for low frequencies. As the frequency increases, a noticeable phase shift begins to occur due to the performance characteristics of the op-amp.

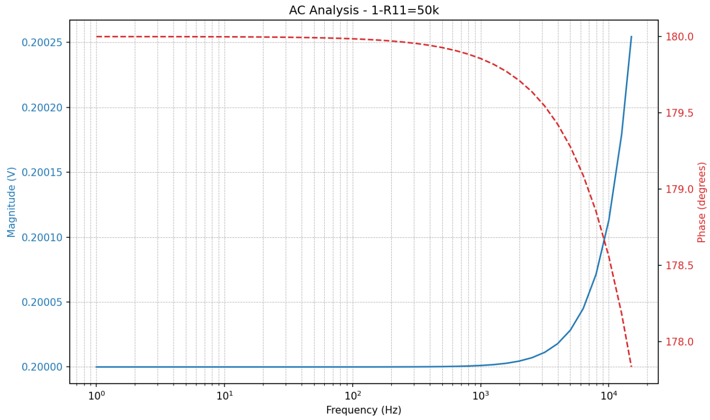


Figure 5: AC analysis results showing the magnitude and phase response for the case where R11 = 50 kΩ.

According to our requirement to keep the phase deviation within 2 degrees (i.e., between 180 and 178 degrees), the plot shows that the phase crosses the 178-degree threshold at a frequency of approximately 13 kHz. Therefore, to ensure the desired signal integrity with minimal phase distortion, the operational frequency for this sensor interface should be maintained below 13 kHz. The output magnitude (blue, solid line) remains relatively constant within this frequency range, confirming that the primary limiting factor for high-frequency operation is the phase shift introduced by the amplifier.

4.7 Measurement Principles and Automated simulation

To determine the individual resistance of a sensor within the matrix, a transient analysis was performed. For this simulation, we focused on a single sensor, R11, as a representative case. The circuit, shown in Figure 6, is stimulated with a sinusoidal voltage input with a frequency of 5 kHz and an amplitude of 0.5V. This input amplitude was carefully chosen to ensure that the amplified output voltage from the operational amplifier remains safely below the hardware-imposed limit of 3.3V.

The core of the measurement technique is to calculate the unknown sensor's resistance (R_{mn}) based on the ratio of the input signal's amplitude to the output signal's amplitude. The calculation is performed using the formula presented below:

$$R_{mn} = R_{ref} \frac{A_m}{B_{mn}} \tag{12}$$

After calculating the "Measured Resistance," we evaluate the accuracy of our method by comparing it against the "Real Resistance" (the actual value set in the simulation). The percentage deviation is calculated using the standard formula shown below:

Performing this analysis manually for a wide range of different sensor resistance values is an exceptionally time-intensive and repetitive task. Each step—changing the resistance parameter, running the simulation, exporting the data, and performing the calculations—would need to be done individually.

To overcome this significant bottleneck, the entire workflow was automated by interfacing LTspice with a Python script. This automated approach allows us to define a range of resistance values to be tested. The script then systematically:

1. Modifies the resistance parameter of the sensor (R11) in the LTspice model.
2. Executes the simulation.
3. Parses the output data to extract the necessary voltage amplitudes.
4. Performs the resistance and deviation calculations using the equations mentioned.
5. Aggregates the results and automatically generates the final plots.

This automation not only saves a considerable amount of time but also enhances the reliability and repeatability of the results by eliminating the potential for manual error. The automated process generates

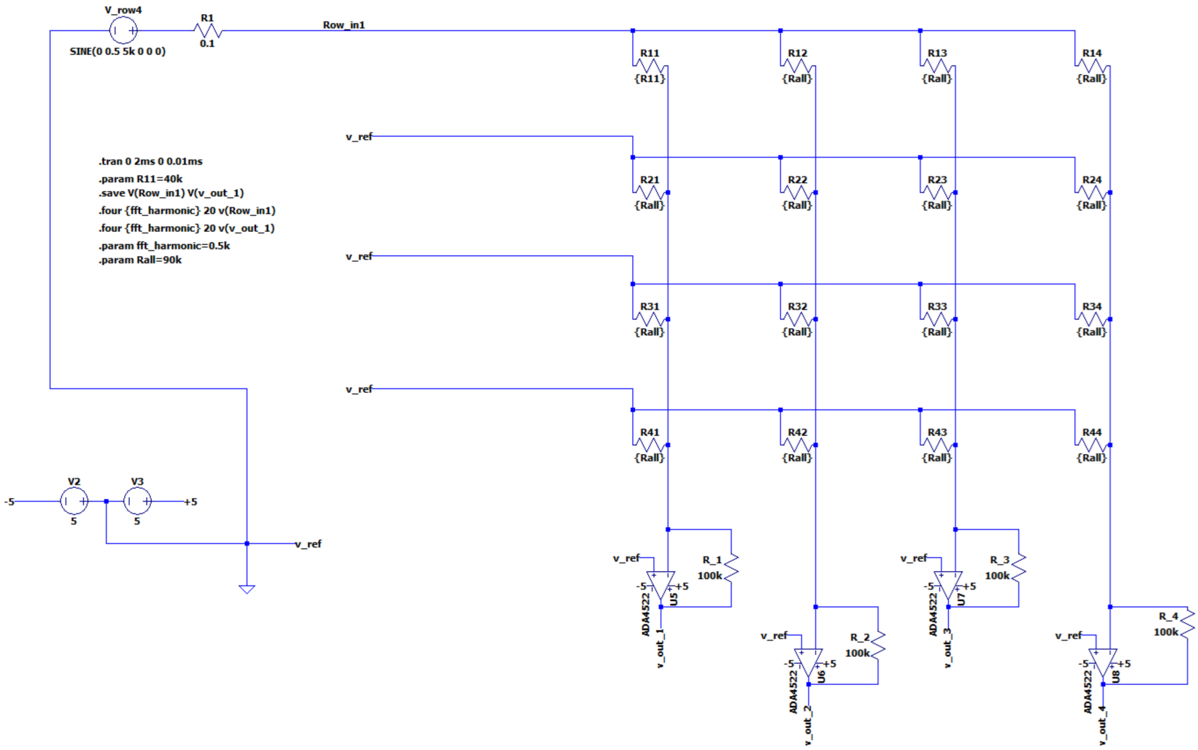


Figure 6: Circuit schematic configured for transient analysis of sensor R11.

$$\text{Deviation} = \frac{\text{Measured Resistance} - \text{Real Resistance}}{\text{Real Resistance}} \times 100\%.$$

detailed results for each simulated resistance value. Figure 7 shows a sample result from a single run where the real resistance of R11 was set to 50 kΩ. The top plot shows the sinusoidal output voltage from the amplifier, and the bottom plot shows its Fast Fourier Transform (FFT). The FFT is crucial as it allows us to precisely measure the magnitude of the output signal (B_{mn}) at the fundamental frequency of 5 kHz.

By running this process over many different values of R11, we can plot the final calculated deviation as a function of the sensor's resistance, as shown in Figure 8.

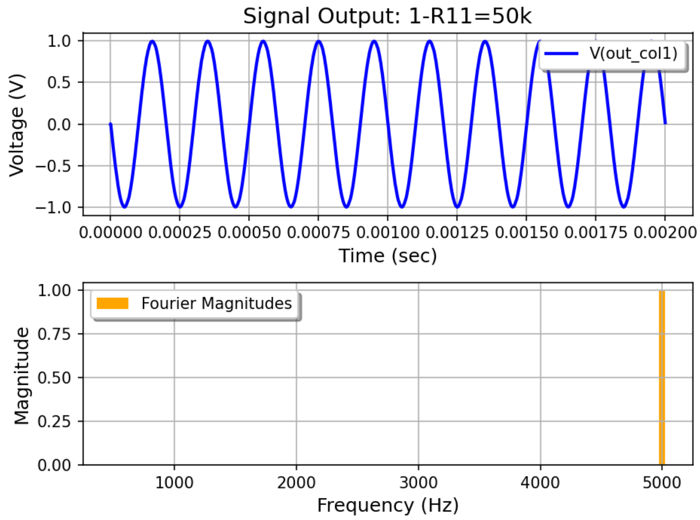


Figure 7: Example of a single simulation run for R11 = 50 kΩ. Top: Time-domain output voltage. Bottom: Fourier analysis showing a clear peak at 5 kHz, used to determine the output amplitude.

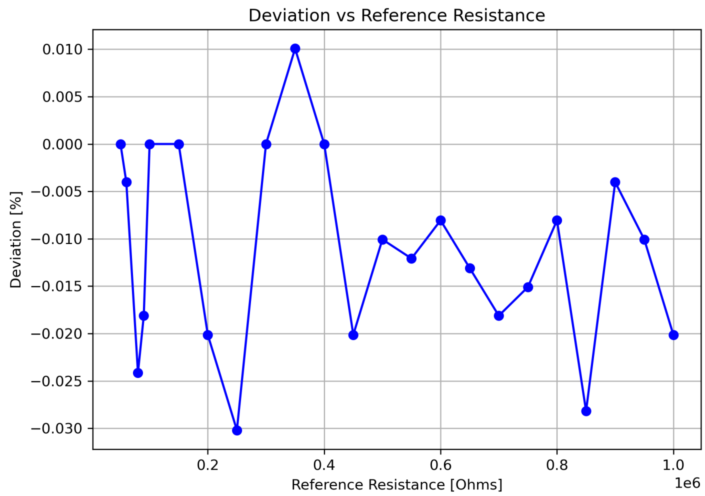


Figure 8: Final calculated deviation as a function of the set sensor resistance (labeled "Reference Resistance" on the axis).

The results in Figure 8 are excellent, demonstrating that the measurement error is extremely low. Across the entire range of tested resistances (from 100 kΩ down to 1 MΩ), the deviation between the measured value and the real value is consistently less than 0.03%. This confirms that the automated simulation and calculation methodology is highly accurate and valid for characterizing the sensor system.

4.7.1 Analysis using Multi-Sine Wave Excitation

To further test the system's performance under more demanding and realistic conditions, the single sine wave input was replaced with a multi-sine wave signal. This approach helps to evaluate the system's ability to accurately measure a specific frequency component in the presence of other signals.

A new input signal is a composite wave created by summing four distinct sine waves with the following frequencies: 500 Hz, 1 kHz, 2.5 kHz, and 5 kHz. Each individual wave maintains an amplitude of 0.5V.

The fundamental principle for calculating the sensor's resistance remains the same as in the single-sine experiment, using the equations previously shown in the last simulation . The key difference in the analysis is that we must first isolate the magnitude of the desired frequency component from the complex output signal using a Fast Fourier Transform (FFT).

The automated simulation process was repeated using this new multi-sine input. Figure 9 shows the result for a single test case where the sensor resistance R11 was set to 50 kΩ.

The time-domain signal (top plot) is now a complex waveform, representing the sum of the system's responses to the four input frequencies. The FFT of this signal (bottom plot) is essential, as it clearly separates and displays the magnitude of each individual frequency component. From this spectrum, we can extract the magnitude of the 5 kHz component to use as the value for B_{mn} in our resistance calculation.

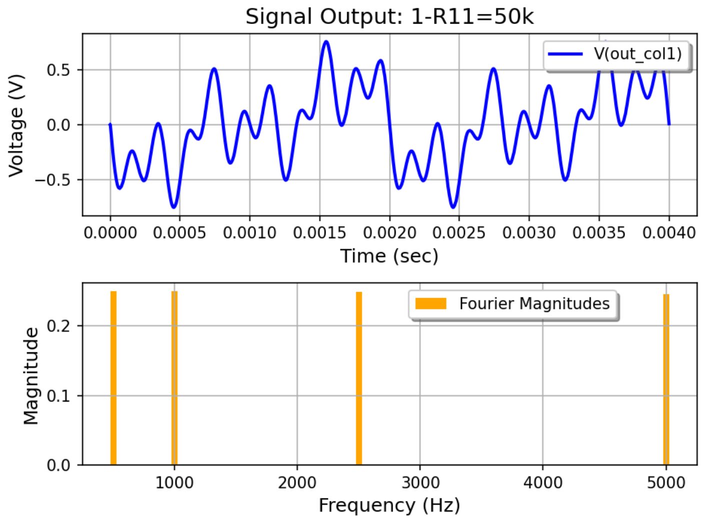


Figure 9: Circuit configured with a multi-sine voltage source for advanced analysis.

The final deviation was calculated for a range of R11 values, and the results are plotted in Figure 10.

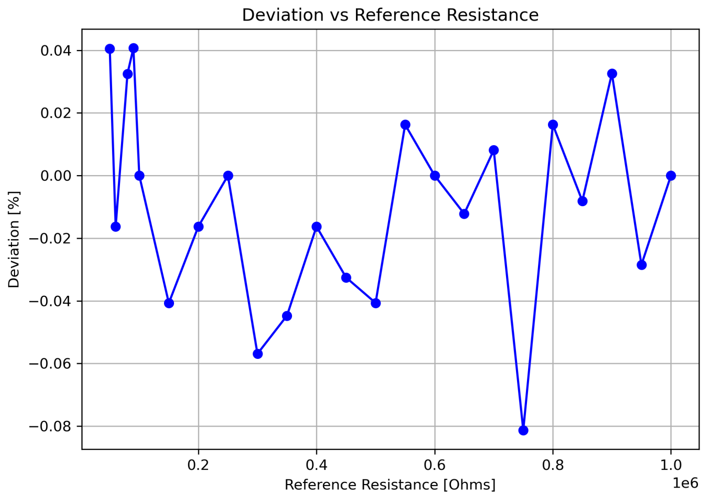


Figure 10: Final calculated deviation versus the set sensor resistance using the multi-sine excitation method.

The results show that even with a more complex input signal, the measurement system remains highly accurate. The percentage deviation is consistently very low, remaining within a band of approximately ±0.08%. While this deviation is slightly larger than that observed in the single-sine wave experiment, it is an expected outcome due to the increased complexity and potential for minor inter modulation effects. Overall, this result strongly validates the robustness of the measurement methodology, confirming its effectiveness even in a more crowded spectral environment.

4.7.2 Analysis using Square Wave Excitation

In the final test configuration, the system's performance was evaluated using a square wave as the input signal. A square wave provides a stringent test for a system because, unlike a pure sine wave, it is a complex signal composed of a fundamental frequency and an infinite series of its odd harmonics (3f, 5f, 7f, etc.). This broadband nature effectively tests the system's response across many frequencies at once.

The method for calculating the sensor's resistance relies on the previous formulas, but it requires careful application. We cannot simply use the peak voltage of the square wave as the input amplitude A_m. Instead, the calculation must be performed in the frequency domain.

According to Fourier analysis, a perfect square wave with a peak amplitude of A_{peak} has a sinusoidal component at its fundamental frequency (f_0) with an amplitude equal to $(4/\pi) \times A_{peak}$. This value is used as the effective input amplitude A_m in our calculation. Correspondingly, a Fast Fourier Transform (FFT) is performed on the amplifier's output signal, and the magnitude of the response at the same fundamental frequency (f_0) is extracted to be used as the measured output amplitude, B_{mn}.

The automated simulation was run using a square wave input with a fundamental frequency of 5 kHz. A sample of the data from a single run, with R11 set to 50 kΩ, is shown in Figure 11.

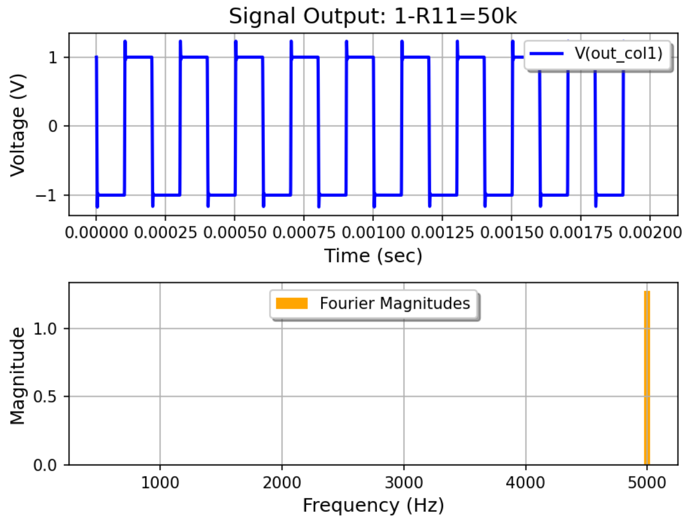


Figure 11: Output signal from the amplifier for a square wave input. The top plot shows the time-domain square wave response. The FFT plot (bottom) isolates the magnitude of the fundamental 5 kHz frequency component, which is the value (B_{mn}) used for the resistance calculation.

The process was repeated for numerous values of the sensor resistance R11 to generate the final deviation plot, as shown in Figure 12.

The analysis shows that the deviation of the measured resistance from the real value is consistently low, generally remaining within a range of ±0.06%. This level of accuracy, achieved with a complex broadband input like a square wave, is a strong testament to the system's robust design. It demonstrates that the amplifier and measurement methodology can effectively handle non-sinusoidal signals by isolating the fundamental frequency, leading to reliable sensor resistance measurements.

4.7.3 Analysis using a Linear Chirp Signal

The final and most comprehensive test was performed using a Linear Chirp Signal (LCS) as the input. An LCS is a signal in which the frequency increases or decreases linearly with time. As seen in the top-left panel of Figure 13, the input signal smoothly sweeps through a range of frequencies. This type of signal is ideal for probing the system's behavior across a continuous frequency band in a single measurement, revealing any frequency-dependent characteristics that discrete frequency tests might miss.

Calculating the resistance with a dynamic signal like a chirp requires an adaptation of our frequency-domain method. Since there is no single stable frequency, the calculation is performed by comparing the spectral density of the input and output signals. Specifically, a Fast Fourier Transform (FFT) is applied to both the input and output time-domain signals, resulting in the frequency-domain plots shown in the right panels of Figure 13. To apply the resistance formula, the magnitudes of the input (A_m) and output

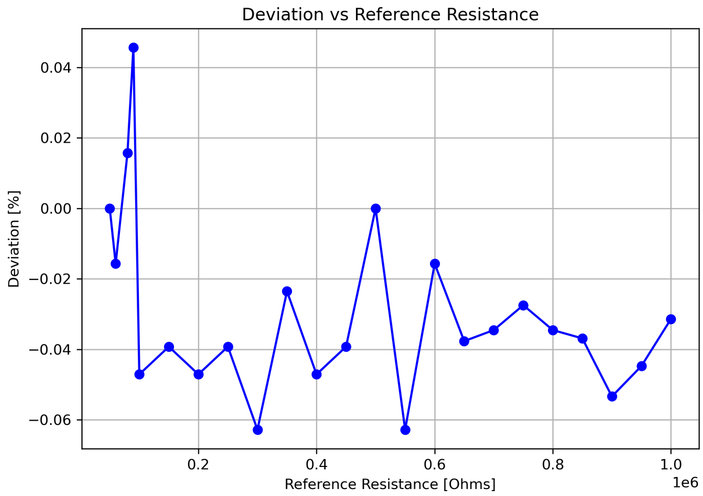


Figure 12: Final calculated percentage deviation as a function of the set sensor resistance, using square wave excitation.

(B_{mn}) are taken at a specific, representative frequency within the chirp's sweep range. This ratio of magnitudes at a single frequency point is then used to calculate the sensor's resistance.

4.7.4 Linear Chirp Simulation Results

The automated simulation was executed with the Linear Chirp Signal input, and the results for a single run are shown in Figure 13.

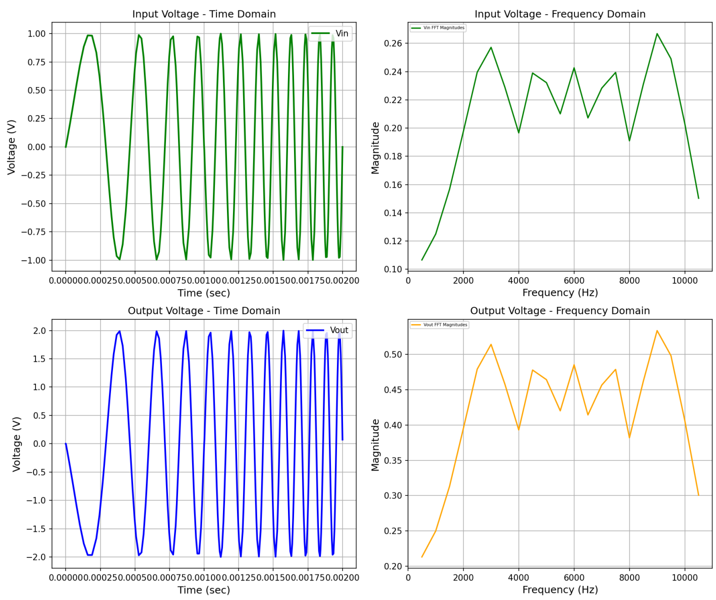


Figure 13: Input and output signals for the Linear Chirp Signal test. Top row: Input signal in the time domain (left) and frequency domain (right). Bottom row: Corresponding amplified output signal in the time and frequency domains.

The automated process was repeated for various values of the sensor resistance R11, and the final percentage deviation was calculated for each run. The aggregated results are plotted in Figure 14.

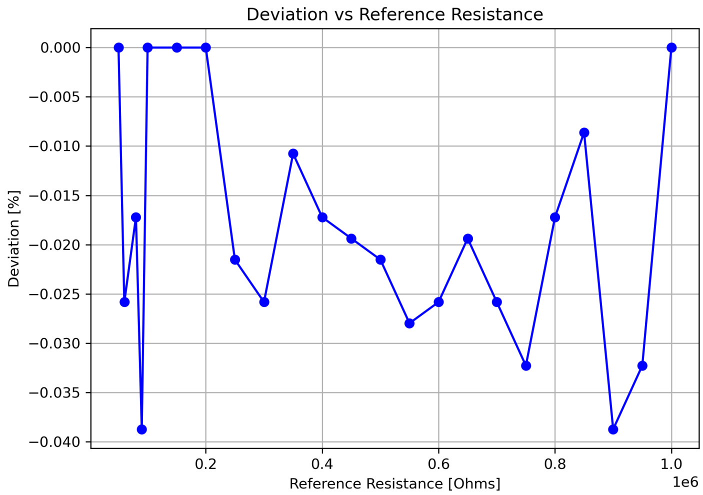


Figure 14: Final calculated percentage deviation as a function of the set sensor resistance, using Linear Chirp Signal excitation.

The final deviation results are presented and the measured deviation remains exceptionally low, falling within a narrow band between 0% and -0.04%. The ability of the system to maintain such high accuracy when stimulated by a complex, time-varying, and broadband chirp signal is the strongest validation of its performance. It confirms that the measurement methodology is robust and reliable across the entire operational frequency band tested in the sweep.

4.7.5 Analysis using a DIBS Input Signal

To assess the system's performance under conditions that mimic modern digital sensing and communication, a final test was conducted using a Delay and Doppler Invariant Binary Sequence (DIBS) as the input signal. A DIBS is an advanced, pseudo-random binary sequence designed to have stable properties, making it useful in sophisticated applications. For this analysis, it serves as a complex, broadband digital-like stimulus.

The time-domain signal, as seen in the top panel of Figure 15, is a non-repeating binary pattern. Its corresponding frequency spectrum, shown in the bottom panel, is composed of multiple distinct frequency components, confirming its broadband nature.

The resistance calculation follows the established frequency-domain methodology. The process involves:

1. Applying the DIBS input signal to the circuit.
2. Performing a Fast Fourier Transform (FFT) on both the input signal and the resulting output signal.
3. Selecting a single, consistent frequency component from the resulting spectra to analyze. For instance, the component at 5000 Hz could be chosen.
4. Using the magnitude of the selected component from the input FFT as A_m and from the output FFT as B_{mn} in the resistance formula.

The automated simulation was executed using the DIBS input. Figure 15 shows the output from the amplifier for a single run where R11 was set to 50 kΩ.

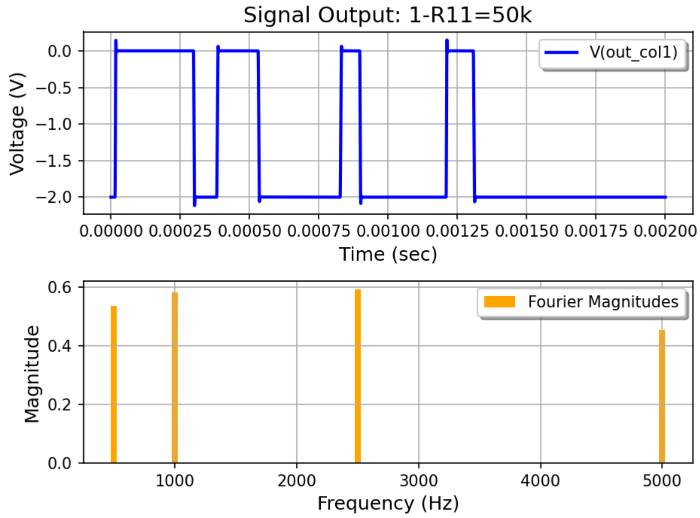


Figure 15: Output signal from the amplifier for a DIBS input. The top plot shows the complex binary sequence in the time domain. The FFT plot (bottom) reveals the broadband, multi-tone nature of the signal's frequency content.

By repeating this process for a range of sensor resistances, the final deviation plot was generated.

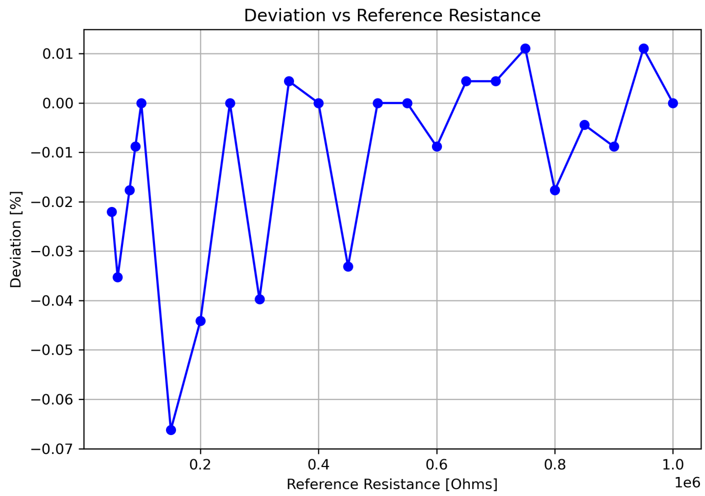


Figure 16: Final calculated percentage deviation as a function of the set sensor resistance, using DIBS excitation.

The deviation results, plotted in Figure 16, demonstrate the system's excellent accuracy and robustness. Even when stimulated with a complex coded sequence like DIBS, the measurement deviation remains extremely low, primarily within a tight band of +0.01% to -0.07%. This successful result confirms that the system is not only reliable for various analog signals but is also well-suited for applications involving more sophisticated, digital-like stimuli. This validates its high performance for a wide range of potential real-world scenarios.

5 Comparison of Excitation Signals

Different excitation signals can be used for impedance spectroscopy, each with distinct characteristics that make them suitable for various applications, where the primary trade-off is often between measurement time, accuracy, and the complexity of the hardware required for signal generation and processing. The conventional frequency-swept sine signal, for example, is very precise but its application can be time-consuming, making it primarily suited for laboratory settings where time is less critical. As the most common alternative, the multisine signal generated by summing multiple sine waves at specific frequencies—is well-suited for applications that require real-time measurements and dynamic responses. Following this is the square signal, a non-sinusoidal waveform that alternates between two fixed levels and is very simple to generate digitally; because it contains a fundamental frequency and all its odd harmonics, it is useful for low-cost systems or for probing multiple frequencies at once, though with less spectral purity than a true multisine signal. A simpler binary version of the multisine, the Discrete Interval Binary Sequence (DIBS), is created by applying a binary threshold to a multisine waveform and is similarly recommended for systems needing to capture real-time responses. Finally, chirp signals, which are defined by a changing frequency over time (such as linear or exponential chirps), are most appropriate for conducting a thorough analysis where the entire frequency bandwidth is of interest.[4]

6 Hardware Implmentation

This project details the development of a real-time embedded system for performing impedance measurements across a two-dimensional sensor matrix. Based on the high-performance STM32H7B0VBTx microcontroller, the system employs the AC Zero Potential Circuit (AC-ZPC) method to enable precise, high-speed impedance imaging, making it ideal for applications in impedance spectroscopy.

6.1 System Architecture and Hardware

The system's core is the STM32H7B0VBTx microcontroller, which orchestrates all signal generation, acquisition, and processing tasks. Key hardware components and their roles are:

- **Signal Injection:** An on-chip Digital-to-Analog Converter (DAC) generates a user-defined AC excitation signal.
- **Signal Acquisition:** An on-chip Analog-to-Digital Converter (ADC) is configured in a dual-mode, regular simultaneous configuration to acquire the output signal from the sensor matrix.
- **Sensor Selection:** Two external analog multiplexers (MUX) are used to dynamically select the target sensor by controlling the row-side signal injection and the column-side sensing path.

To ensure signal integrity and accurate measurements of a single sensor within the matrix, a specific process is implemented. First, a user-defined AC signal is injected into a single, selected row in a step known as targeted excitation. To prevent interference from adjacent sensors and minimize crosstalk, only the target sensor is excited at any given time. The output signal from the corresponding selected column is then passed through a voltage buffer before being fed into the ADC, which minimizes loading effects and preserves signal integrity. Finally, the excitation and acquisition processes are precisely synchronized using the microcontroller's advanced timers and Direct Memory Access (DMA) controller to ensure accurate sampling with minimal CPU intervention.

6.2 Real-Time Data Processing and Analysis Resullt

Once a complete frame of sensor data is acquired, the system performs all calculations in real-time. The processing pipeline is as follows:

1. **Data Ready Flag:** The completion of an ADC conversion frame is signaled by setting a flag (*ConvCmpltFlag*), triggering the data processing routine.
2. **FFT Computation:** A Fast Fourier Transform (FFT) is executed on the captured voltage and current signal data. This task leverages the powerful CMSIS-DSP library, which is hardware-accelerated by the STM32H7's Floating-Point Unit (FPU) and high clock speed.
3. **Component Extraction:** The real and imaginary components of both voltage and current are extracted from the specific frequency bin corresponding to the initial excitation frequency.
4. **Impedance Calculation:** Using the extracted complex components, the system computes the magnitude and phase of the voltage and current to determine the sensor's impedance ($Z = V/I$).

To enhance accuracy, the raw impedance value is corrected using a polynomial calibration function. This function is adapted from the theoretical model of the AC-ZPC method and effectively compensates for non-idealities in the operational amplifiers and other parasitic effects within the circuit.

Here is a table showing the true impedance values alongside other placeholder data.

Measured Impedence (Ω)	True Impedence (Ω)	Relative Deviation(Measured/True)	Measured Phase	Absolute Phase Error (from -355°)
69359	68000	1.019676471	-355	0
77461	75000	1.033706667	-355	0
99746	100000	0.99746	-355	0
221561	220000	1.00709	-355	0
469991	470000	0.9999	-355	0

6.3 System Capabilities and Data Interface

The final corrected impedance magnitude and phase values are stored in a 2D array structure that mirrors the physical sensor matrix. For visualization and further analysis, the complete impedance map is then transmitted via UART to a host PC or data logger. The system enables full impedance imaging of the entire sensor matrix by supporting multi-point scanning, which is achieved by iteratively changing the row and column selections via the multiplexers.

This implementation successfully demonstrates a high-performance solution for real-time impedance spectroscopy, where the choice of the STM32H7B0VBTx microcontroller was critical to the project's success. This microcontroller offered key features essential for the application, including high-speed, synchronized ADC and DAC interfaces; advanced timer and DMA capabilities for precise, low-overhead operation; and sufficient on-chip RAM to handle the large data buffers required for FFTs. Furthermore, its hardware-accelerated DSP instructions and a Floating-Point Unit (FPU) enabled the efficient real-time computation necessary for the system's performance.

7 Conclusion

It has successfully addressed the critical challenge of crosstalk in 2-D sensor arrays by thoroughly investigating and validating the Alternating Current Zero Potential Circuit (AC-ZPC) methodology. The work progressed from a theoretical framework to a fully realized hardware prototype, demonstrating a comprehensive and effective engineering solution.

The extensive automated simulations were pivotal, establishing not only the operational limits of the proposed circuit but also confirming its exceptional robustness. By testing the system against a diverse range of excitation signals—from simple sinusoids to complex, broadband inputs like chirp and DIBS—the measurement technique's resilience and high fidelity were unequivocally proven. The consistently low deviation percentages achieved across all test cases are not merely data points; they are a powerful testament to the validity of the frequency-domain analysis approach and its ability to reject noise and interference, thus confirming its suitability for high-precision applications.

Ultimately, the successful implementation on the STM32H7B0VBTx platform transformed the theoretical concept into a tangible, high-performance instrument. This transition from simulation to a real-world system capable of real-time impedance spectroscopy underscores the project's primary achievement: the development of a reliable, accurate, and versatile measurement system. The findings confirm that the AC-ZPC method, when combined with modern microcontrollers and robust data processing, provides a superior solution for advanced sensor imaging, opening possibilities for its use in demanding industrial and scientific fields.

8 References

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